



REDEFINING  
MARITIME  
SUPERIORITY\_

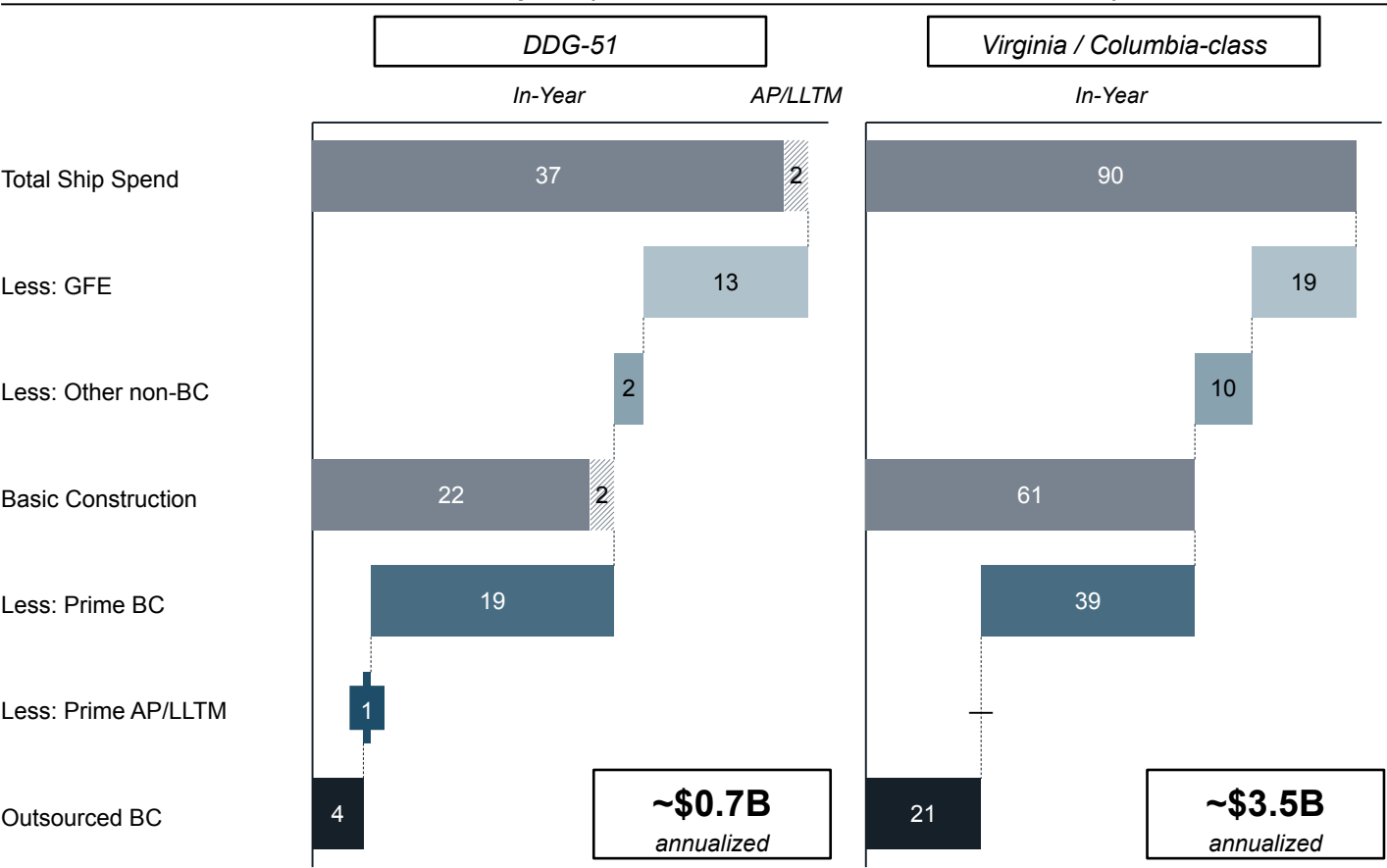
## Distributed Shipbuilding Supplier Market

*DDG and submarine new-construction component suppliers, FY2022–FY2027*

June 2026

# Outsourced Basic Construction | Total ship spend narrows to ~\$0.7B of outsourced work per year for DDG-51 and ~\$3.5B for submarines, FY2022–FY2027.

Walk to Outsourced Basic Construction Spend (\$B, cumulative FY2022–FY2027, FY2026 \$)

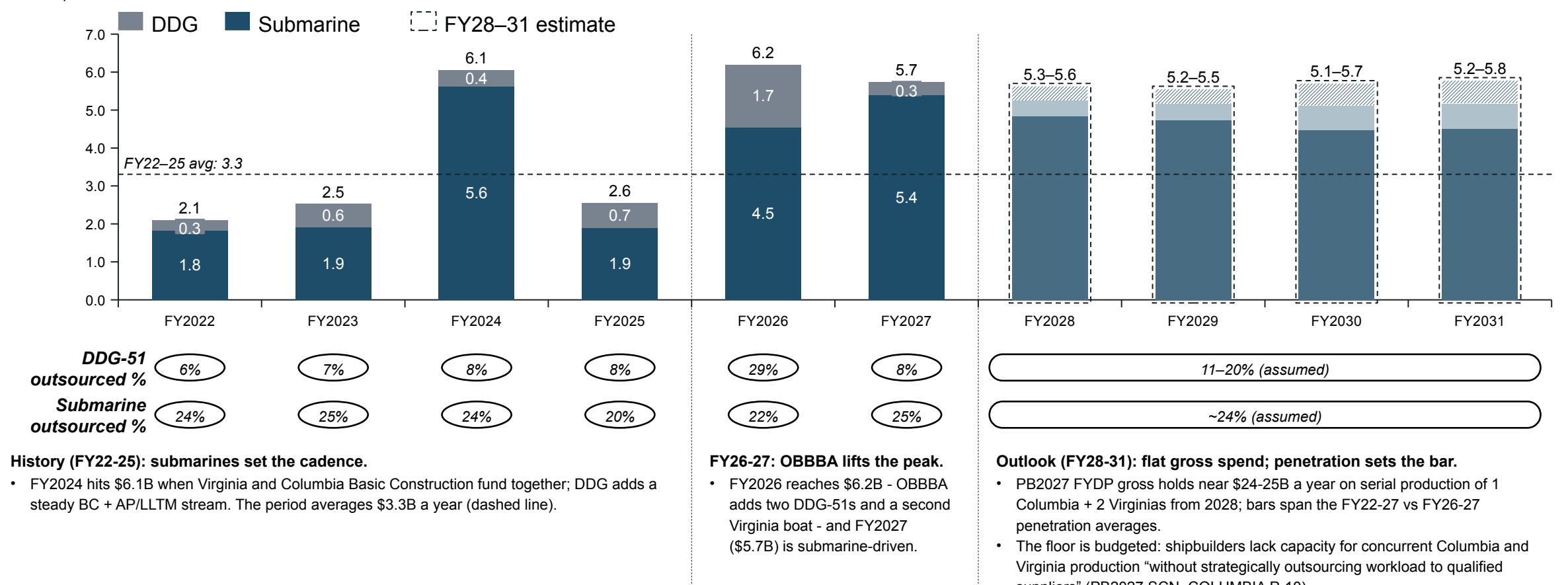


| Step                | Rationale                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Ship Spend    | <ul style="list-style-type: none"><li>FY2022–FY2027 SCN end cost (P-5c discretionary) plus OBBBA Sec. 20002 mandatory awards.</li><li>OBBBA adds two FY2026 DDG-51s (\$5.4B) and a second FY2026 Virginia boat (\$4.6B).</li><li>DDG-51 adds incremental FY2026 AP/LLTM (\$1.8B); submarines add none (supplier long-lead is already inside Basic Construction).</li></ul> |
| Less: GFE           | <ul style="list-style-type: none"><li>Removes government-furnished equipment: propulsion, electronics, and ordnance the Navy buys directly and furnishes to the yard.</li><li>Includes the non-BC remainder of OBBBA awards.</li></ul>                                                                                                                                     |
| Less: Other non-BC  | <ul style="list-style-type: none"><li>Removes plans, change orders, and other non-construction categories.</li><li>Engineering and program cost, not ship work.</li></ul>                                                                                                                                                                                                  |
| Basic Construction  | Calculation                                                                                                                                                                                                                                                                                                                                                                |
| Less: Prime BC      | <ul style="list-style-type: none"><li>Removes construction the prime yards self-perform.</li><li>Outside-yard share from FPDS subaward place-of-performance evidence: 12.5% for DDG-51 (MYP-corrected), 35.0% for submarines.</li></ul>                                                                                                                                    |
| Less: Prime AP/LLTM | <ul style="list-style-type: none"><li>Removes AP/LLTM retained by the primes: 80% ship-construction share x 85% supplier-addressable, applied to the DDG-51 FY2026 AP base.</li><li>Submarines: none.</li></ul>                                                                                                                                                            |
| Outsourced BC       | Calculation                                                                                                                                                                                                                                                                                                                                                                |

Sources: (1) Navy SCN/P-5c budget exhibits, FY2022–FY2027 (DDG-51; Virginia and Columbia classes); (2) OSD FY2025 Green Book procurement deflators (constant FY2026 \$); (3) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002; (4) FPDS/FSRS subaward place-of-performance evidence (outside-yard shares)

# Outsourced Basic Construction (Annual TAM) | Demand steps up from a \$3.3B pre-OBBBA average to a \$6.2B peak in FY2026, and is expected to hold \$5.1–5.8B a year through FY2031.

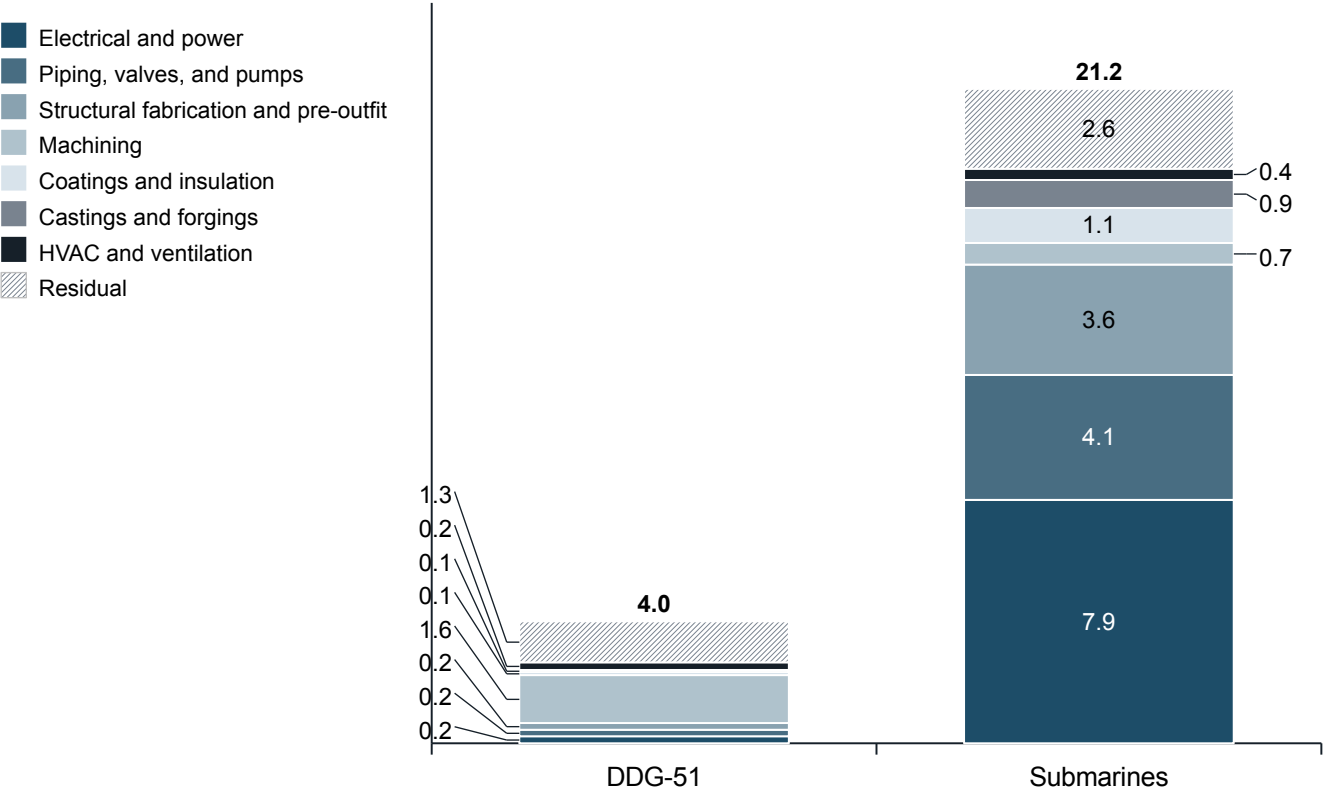
Outsourced Basic Construction TAM by fiscal year, \$B; stacked by program; constant FY2026 dollars. FY2022–FY2027 incl. OBBBA Sec. 20002; FY2028–FY2031 estimated (hatched range, dashed frames): PB2027 FYDP gross × historical penetration.



Sources: (1) Navy SCN P-5c / P-40 budget justification, FY2022–FY2027, and PB2027 FYDP outyears (FY2028–FY2031); (2) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002; (3) PB2027 SCN, COLUMBIA P-10 Strategic Outsourcing narrative; (4) OUSD(C) Green Book Procurement deflators. FY2028–31 = PB2027 FYDP gross × penetration; range spans the FY22–27 vs FY26–27 average penetration; constant FY2026 \$.

# Outsourced Basic Construction Spend by Work Type | Electrical power leads the ~\$21.2B submarine pool; machining leads the ~\$4.0B DDG-51 pool.

Outsourced Basic Construction Spend by Work Type (\$B, cumulative FY2022–FY2027, FY2026 \$)



Methodology

| Stage                  | Rule applied                                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Award pull          | Pull FFATA/FSRS subawards; collect visible supplier evidence (mix, not TAM).                                       |
| 2. PIID scope          | Scope each award to DDG or submarine by parent-prime PIID.                                                         |
| 3. Entity resolution   | Resolve the operating entity paid (UEI), not the parent brand.                                                     |
| 4. Role filter         | Drop non-supplier / non-component roles from the addressable base.                                                 |
| 5. Work-type assign    | First clean match wins one work-type home (registry, override, NAICS-4).                                           |
| 6. Residual discipline | Keep unresolved dollars in the addressable base; residual dilutes named shares.                                    |
| 7. Share output        | Bucket dollars over the addressable base set the modeled work-type and residual shares; feeds the allocation step. |

The classifier is entity-first and role-first.

- PIID scopes the award; UEI resolves the entity paid; roles are filtered before any work-type call.

Residual is unclassified award spend.

- Unresolved dollars stay in TAM but outside SAM: the hatched cap on each program's stack.

Sources: (1) Navy SCN/P-5c budget exhibits, FY2022–FY2027 (DDG-51; Virginia and Columbia classes); (2) OSD FY2025 Green Book procurement deflators (constant FY2026 \$); (3) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002; (4) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026 (work-type shares)

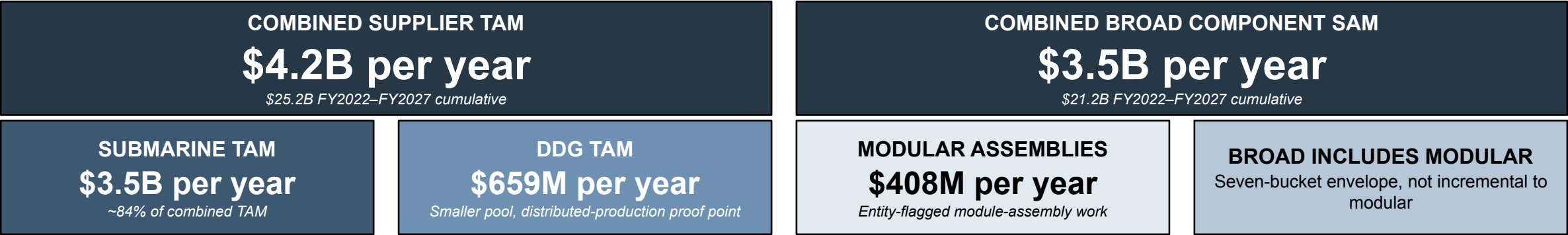
# Market Landscape

*The opportunity is the component layer yards buy out, and demand is funded*

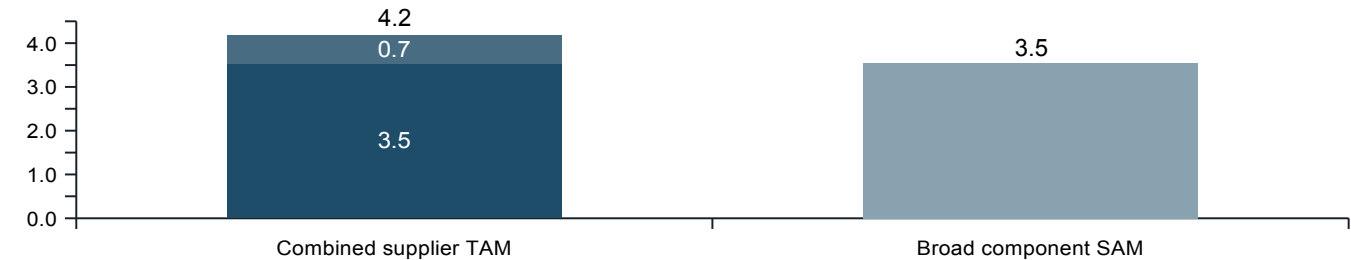


# Supplier TAM and SAM | Combined supplier opportunity is ~\$4.2B per year, with ~\$3.5B addressable as broad component SAM.

FY2022–FY2027 model-period averages, annualized; constant FY2026 dollars; includes OBBBA Sec. 20002 mandatory funding.



Combined supplier TAM stacked by program vs broad component SAM, \$B per year; FY2022–FY2027 average, constant FY2026 dollars.



■ Submarine   ■ DDG   ■ Broad component SAM

**The pool is submarine-led, while DDG supplies the distributed-production proof point.**

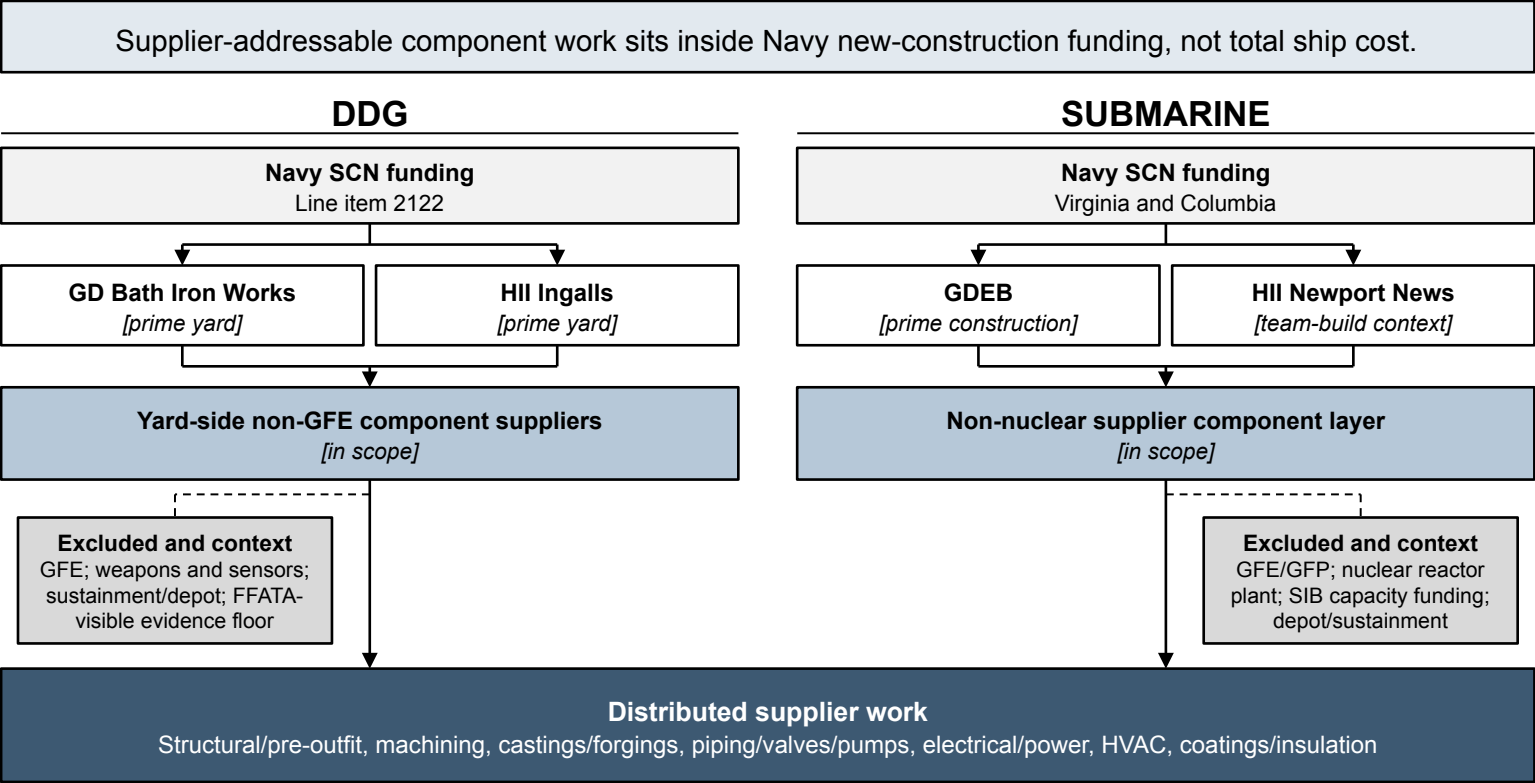
- Virginia/Columbia contributes \$3.5B of the \$4.2B annual TAM; DDG adds \$659M per year and carries the clearest public distributed-production signal.

**Broad SAM is the serviceable envelope, not a capture forecast.**

- Broad component manufacturing is \$3.5B per year across the seven named buckets; modular assemblies is a narrower \$408M entity-flagged cut inside that envelope.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DoD place-of-performance award corpus and parent-prime PIID scope; (3) FFATA/FSRS subaward records and the operating-entity supplier registry; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002

Supplier Layer | The opportunity sits between Navy budget authority and prime-yard construction, in component work yards buy out rather than self-perform.



The primes differ, but the serviceable cut is the same.

- DDG dollars run through Bath Iron Works and Ingalls; submarine dollars run through Electric Boat and Newport News. In both lanes, the counted layer is supplier component work outside GFE, sustainment, and yard self-perform.

Visible subawards prove the layer, not its full size.

- FFATA-visible filings provide supplier names and work-type cues; first-tier visibility thins below the first tier and differs by yard.

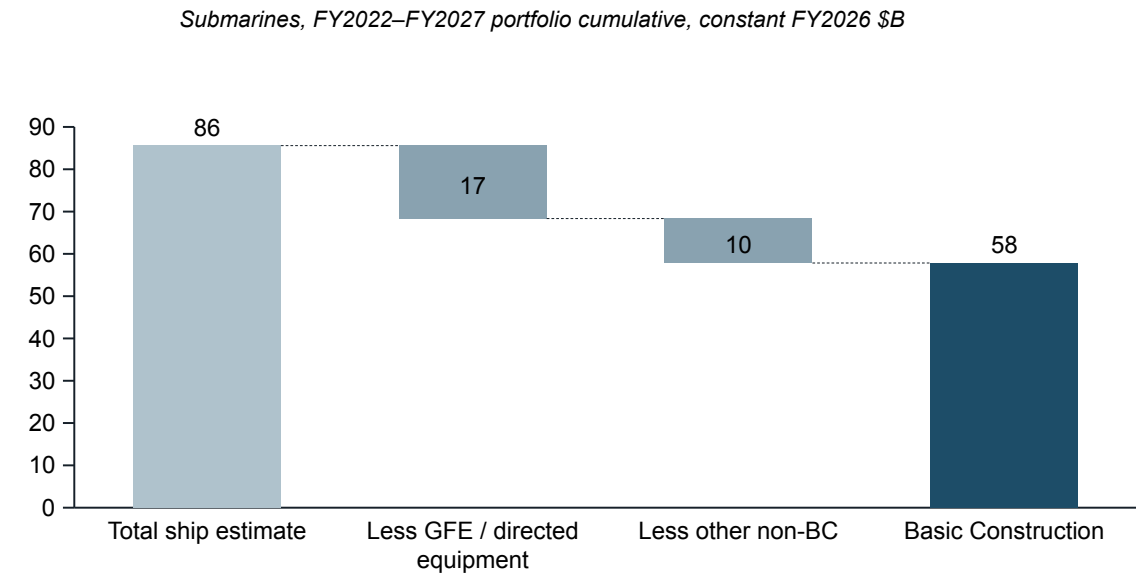
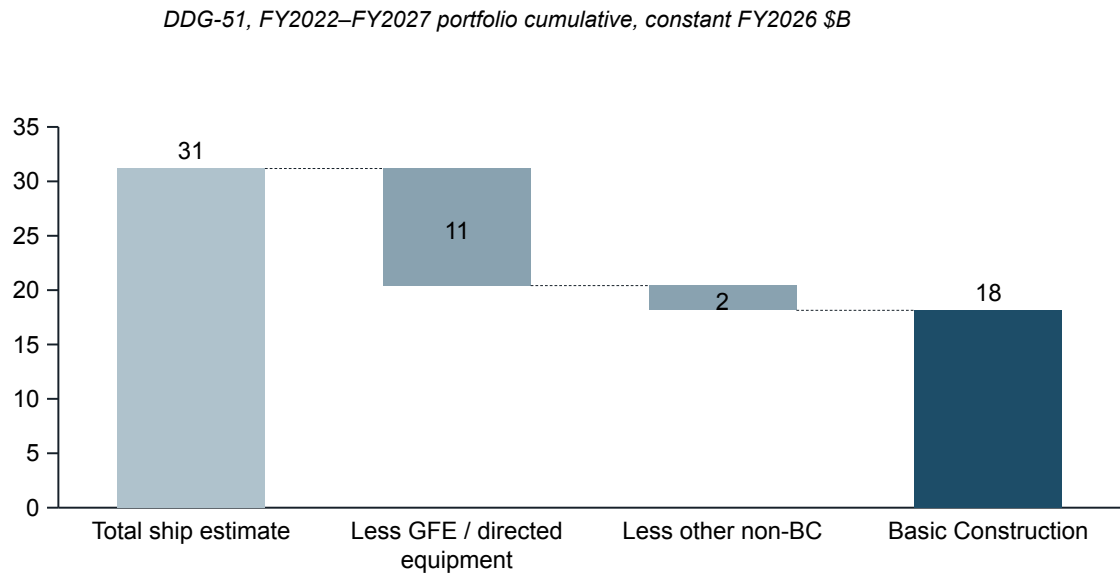
Electrical and power means ship power, distribution, and generation; combat and mission electronics leave the boundary before bucket assignment.

Blue = counted supplier layer; gray = excluded or context only; outline = prime or team-build layer.

Sources: (1) Navy SCN shipbuilding budget justification and line-item structure, FY2022–FY2027; (2) prime-yard place-of-performance award corpus; (3) FFATA/FSRS first-tier subaward records



# Scope and Cost Funnel | Sizing starts at new-construction budget lines and narrows to non-GFE Basic Construction, not total ship cost.



**Basic Construction is the included denominator.**

- The sizing base is \$18.2B of DDG's \$31.2B FY2022–FY2027 estimate and \$57.9B of the submarine portfolio's \$85.6B. This base funds yard labor, team-build work, purchased material, and first- and lower-tier supplier flow.

**GFE, other budget categories, and yard self-perform leave the boundary.**

- GFE leaves first — DDG electronics and ordnance (\$10.8B); submarine government-furnished propulsion, electronics, and ordnance (\$17.2B). Then other non-BC categories — plans, change orders, HM&E, and other, plus Virginia technology insertion on the submarine side (\$2.3B DDG, \$10.5B sub). Yard self-perform within Basic Construction is removed later, at the supplier step.

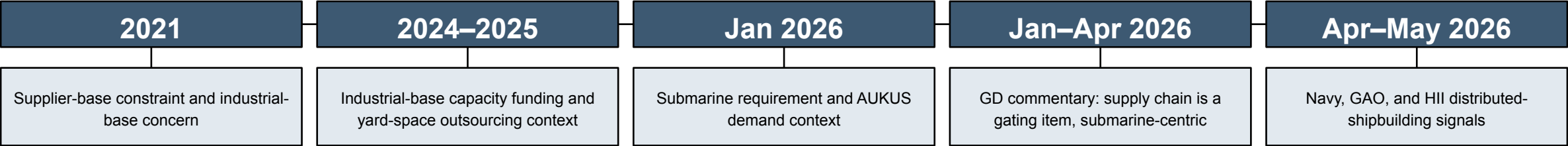
**Top-of-funnel context and first-tier filings are evidence, not numerator.**

- Total ship estimate and TOA frame the denominator; FFATA/FSRS subawards are a visible floor behind the supplier coefficient. Do not add AP/LLTM on top of the BC base inside this funnel; the DDG-only filtered AP/LLTM stream is handled separately, and the OBBBA Sec. 20002 mandatory awards are additive to these P-5c funnels, entering at the TAM bridge.

Sources: (1) DDG and submarine SCN/P-5c cost-funnel data, FY2022–FY2027; (2) Navy P-5c GFE and Basic Construction breakouts; (3) program TAM bridge inputs; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002



# Demand Backdrop | Policy, prime behavior, and industrial-base pressure point toward more distributed supplier capacity.



**Supplier-base constraint:** capacity, long-lead items, sole-source exposure, and industrial-base atrophy.

**Policy and funding tailwind:** Navy distributed-production direction and capacity investment.

**Prime behavior:** HII public distributed-work commitments; GD supply-chain pressure signals.

**Demand signals support more external capacity, but not a larger sizing numerator.**

- Policy, oversight, and prime commentary point toward more distributed supplier work; the FY2022–FY2027 model still sizes from SCN Basic Construction and strict supplier coefficients.

**DDG and submarine signals support the same direction for different reasons.**

- HII carries the clearer DDG distributed-production signal; GD's public signal is more submarine-centric, with margin pressure consistent with an outsourcing premium.



Virginia-class submarine under construction

The model sizes the current opportunity; future distributed work for new entrants is a diligence question, not a modeled input.

Sources: (1) Navy and OSD shipbuilding and industrial-base policy and budget materials, FY2021–FY2026; (2) GAO shipbuilding and submarine industrial-base reports; (3) HII and General Dynamics public disclosures and earnings commentary, 2024–2026; (4) submarine construction photograph, illustrative



# Sizing the Opportunity

*A budget-led TAM with an allocation-led SAM*



# Methodology | TAM is budget-led, while SAM mix is allocated through one entity-resolved supplier registry.

## Sizing chain and governing rules

| LAYER                   | GOVERNING SIGNAL                                                | TREATMENT                                                                 | WHY IT MATTERS                                     |
|-------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------|
| Budget TAM              | SCN Basic Construction and AP/LLTM stream                       | Build exogenous TAM by program                                            | Classification does not move headline TAM          |
| Supplier share          | POP corpus                                                      | Strip prime, co-prime, GFE, and excluded scope                            | Converts the budget base into supplier opportunity |
| Supplier mix            | Operating entity, SAM-resolved supplier registry                | Role first, then bucket; subawards are mix evidence, not market size      | One registry classifies DDG and submarine alike    |
| Physical buckets        | Seven stable work-type buckets                                  | Structural, machining, castings, piping, electrical power, HVAC, coatings | Keeps the two programs comparable                  |
| Exclusions and residual | Mission systems, service, holding, foreign/FMS, unresolved tail | Route out, or hold residual explicit                                      | Broad SAM does not auto-equal TAM                  |
| Scenario SAM            | Bucket and entity flags                                         | Overlapping cuts: metal, HM&E, electrical, modular, broad                 | Scenarios are not additive                         |

- TAM is fixed by the budget method, not by supplier reclassification.**
- Supplier TAM is the budget base multiplied by the place-of-performance supplier share; re-bucketing suppliers changes mix, not the \$4.2B headline.
- The same registry governs both programs after role exclusions.**
- DDG and submarine dollars are classified at the operating-entity level after mission systems, services, holding companies, foreign/FMS, primes, and GFE are removed.
- Residual is a modeled result, not a cleanup item.**
- Unresolved supplier dollars stay in the denominator and dilute bucket shares, so broad SAM does not automatically equal total TAM.

**Modeling boundary:** electrical means ship power, distribution, and generation; combat and mission electronics are excluded in the base case. DDG VLS launch-control is sensitivity, not base scope.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DoD place-of-performance (POP) award corpus and parent-prime PIID scope; (3) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026

# TAM Bridge | Combined supplier TAM is \$4.2B per year, but AP/LLTM is additive for DDG and reference-only for submarine.

Supplier TAM bridge, \$B cumulative and \$M per year; constant FY2026 dollars, incl. OBBBA Sec. 20002 mandatory funding. Each program is bridged separately, then summed.

## DDG supplier TAM bridge

|                                                                         |
|-------------------------------------------------------------------------|
| BC construction base                                                    |
| <b>\$21.5B cumulative</b> incl. \$3.4B OBBBA mandatory (Sec. 20002(17)) |
| Less prime, co-prime, and GFE place-of-performance                      |
| <b>\$18.8B removed</b> scope removal                                    |
| BC supplier stream                                                      |
| <b>\$2.7B cumulative</b> \$451M per year                                |
| Add AP/LLTM stream                                                      |
| <b>\$1.2B cumulative</b> \$208M per year                                |
| DDG supplier TAM                                                        |
| <b>\$4.0B cumulative</b> \$659M per year                                |

## Submarine supplier TAM bridge

|                                                                         |
|-------------------------------------------------------------------------|
| Basic Construction base                                                 |
| <b>\$60.6B cumulative</b> incl. \$2.7B OBBBA mandatory (Sec. 20002(16)) |
| Applied supplier share                                                  |
| <b>35%</b> strict, non-nuclear, yard-excluded                           |
| Submarine supplier TAM                                                  |
| <b>\$21.2B cumulative</b> \$3.5B per year                               |
| Additive AP/LLTM base                                                   |
| <b>\$0</b> supplier long-lead already inside Basic Construction         |

COMBINED SUPPLIER TAM

**\$4.2B per year** \$25.2B cumulative

### DDG has two additive supplier streams.

- Basic Construction supplier work is \$451M per year and filtered AP/LLTM adds \$208M per year, because DDG Basic Construction is net of prior-year AP.

### Submarine AP/LLTM stays reference-only.

- The 35% supplier share on \$60.6B of Basic Construction yields \$3.5B per year; supplier long-lead material already sits inside the boundary, so additive AP/LLTM remains \$0.

### The combined TAM sums dollars, not rules.

- DDG and submarine outputs sum to \$4.2B per year and \$25.2B cumulative; the AP/LLTM treatment remains program-specific and should not be blended.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) DDG MYP-corrected outside-yards place-of-performance evidence; (3) submarine coefficient and AP/LLTM boundary analysis; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002

Supplier Share | Strict inputs hold TAM conservative, but DDG and submarine solve different evidence problems.

DDG APPLIED BC SUPPLIER SHARE

13%

MYP-corrected non-GFE BC stream

SUBMARINE APPLIED BC SUPPLIER SHARE

35%

Strict non-nuclear Basic Construction view

| PROGRAM   | EVIDENCE ISSUE | APPLIED VALUE                                           | BROADER EVIDENCE VIEWS                                                 | FEEDS HEADLINE TAM                        |
|-----------|----------------|---------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------|
| DDG       | MYP redaction  | 13% BC supplier share                                   | Restored \$14.6B multiyear masters; corrected outside-yards POP ~32.8% | 13% on the non-GFE BC stream              |
|           | AP/LLTM        | 85% supplier-share input on 80% ship-construction share | Assumption-driven filtered stream                                      | Additive filtered stream, \$208M per year |
| Submarine | POP breadth    | 35% BC supplier share                                   | All-gated POP anchor ~52%; AP/LLTM reference ~49%                      | 35% on non-nuclear BC                     |
|           | AP/LLTM        | \$0 additive base                                       | Gross AP is large supplier evidence                                    | Not additive under the current boundary   |

DDG supplier-share evidence starts with a redaction correction.

- The model restores \$14.6B of redacted multiyear masters to correct the outside-yards POP read; the applied BC supplier share remains 13%.

Submarine supplier-share evidence is broad but deliberately constrained.

- The 35% input is a strict non-nuclear, yard-excluded view; broader POP anchors near 52% and 49% stay sensitivity.

Only the applied shares feed headline TAM.

- Broader POP views support the audit record, but 13% for DDG and 35% for submarine are the TAM bridge inputs.

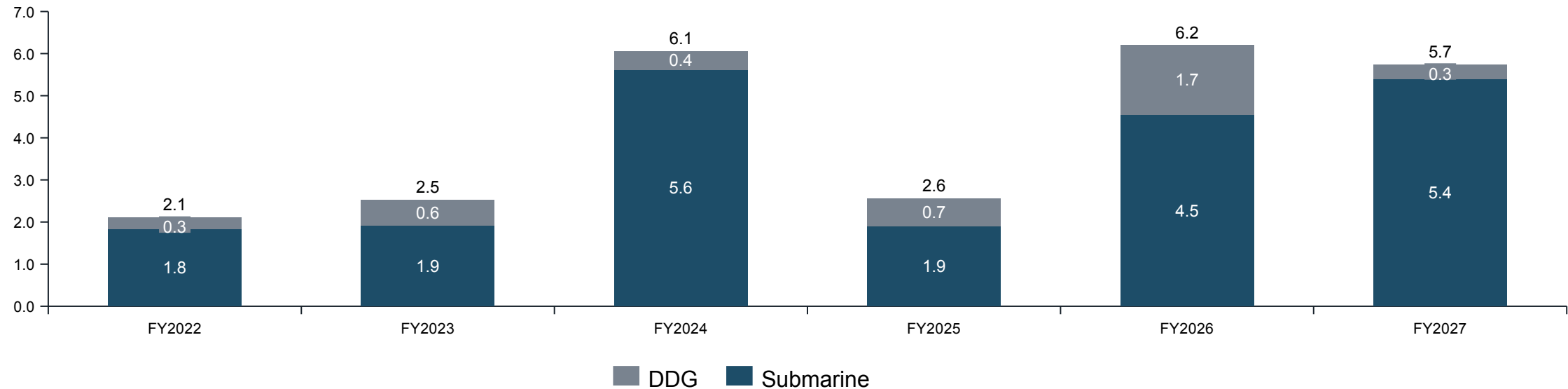
The two programs' shares are inputs to separate models; a single blended supplier-share percentage would mislead.

Sources: (1) DDG MYP redaction correction and outside-yards place-of-performance evidence; (2) submarine coefficient and AP/LLTM boundary evidence; (3) Navy SCN and P-5c Basic Construction budget justification, FY2022–FY2027



# Annual Cadence | Average annual TAM is a sizing convention; actual supplier demand is lumpy and peaks in FY2024 and in OBBBA-boosted FY2026.

Combined supplier TAM by fiscal year, \$B; stacked by program. Constant FY2026 dollars, FY2022–FY2027; incl. OBBBA Sec. 20002 mandatory funding.



**Submarine procurement drives the cadence baseline.**

- FY2024 reaches \$6.1B and FY2027 \$5.7B when Virginia and Columbia Basic Construction fund together.

**FY2026 is the OBBBA and long-lead spike.**

- The combined year reaches \$6.2B: OBBBA mandatory funding adds two DDG-51s and a second Virginia boat, while DDG AP/LLTM long-lead peaks at ~\$1.2B in the same year.

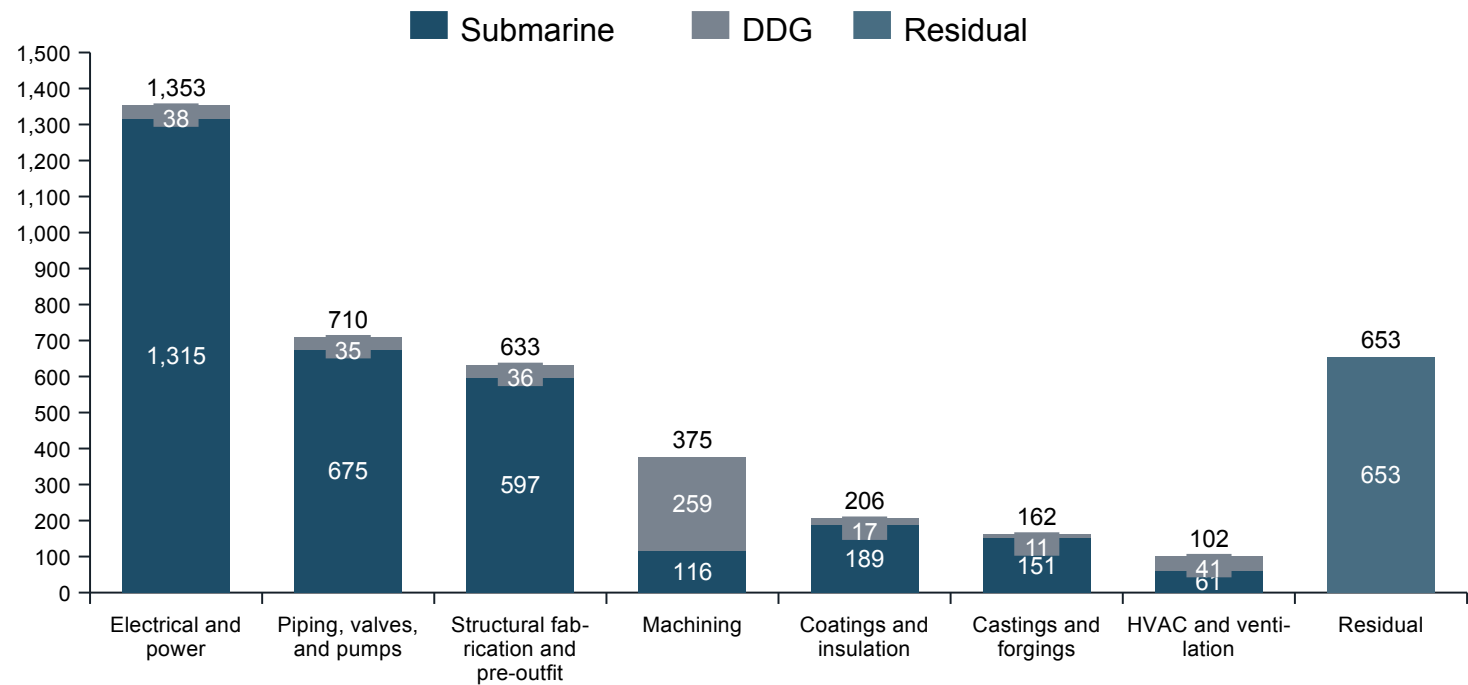
**The \$4.2B average is a sizing convention.**

- The six-year average smooths a path from \$2.1B in FY2022 to \$6.2B in FY2026; the annual shape reflects procurement cadence, not a steady run-rate.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) Virginia and Columbia annual procurement profiles; (3) DoD contract announcements and FFATA/FSRS subaward records; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002

# Work-Type Mix | Electrical power, piping, and structural fabrication lead the combined pool; a \$653M residual stays separate.

Combined average annual TAM by work type, \$M; stacked by program. Residual shown separate. Constant FY2026 dollars.



- Three buckets dominate the named supplier pool.**
- Electrical power, piping, and structural fabrication total ~\$2.7B of the \$4.2B TAM, led by \$1,353M of electrical and power work.
- Entity-resolved classification shifts the mix, not the TAM headline.**
- Machining rises to \$375M once VLS launcher-cell machining and LM2500 propulsion machinery are assigned to physical work type; combat electronics leave the supplier base.
- Residual remains visible outside named-bucket SAM.**
- \$653M of supplier dollars lacks recoverable bucket evidence, so it stays in TAM while remaining outside broad SAM.

The component pool is not one thing; entry, qualification, and incumbent dynamics differ sharply by work type.

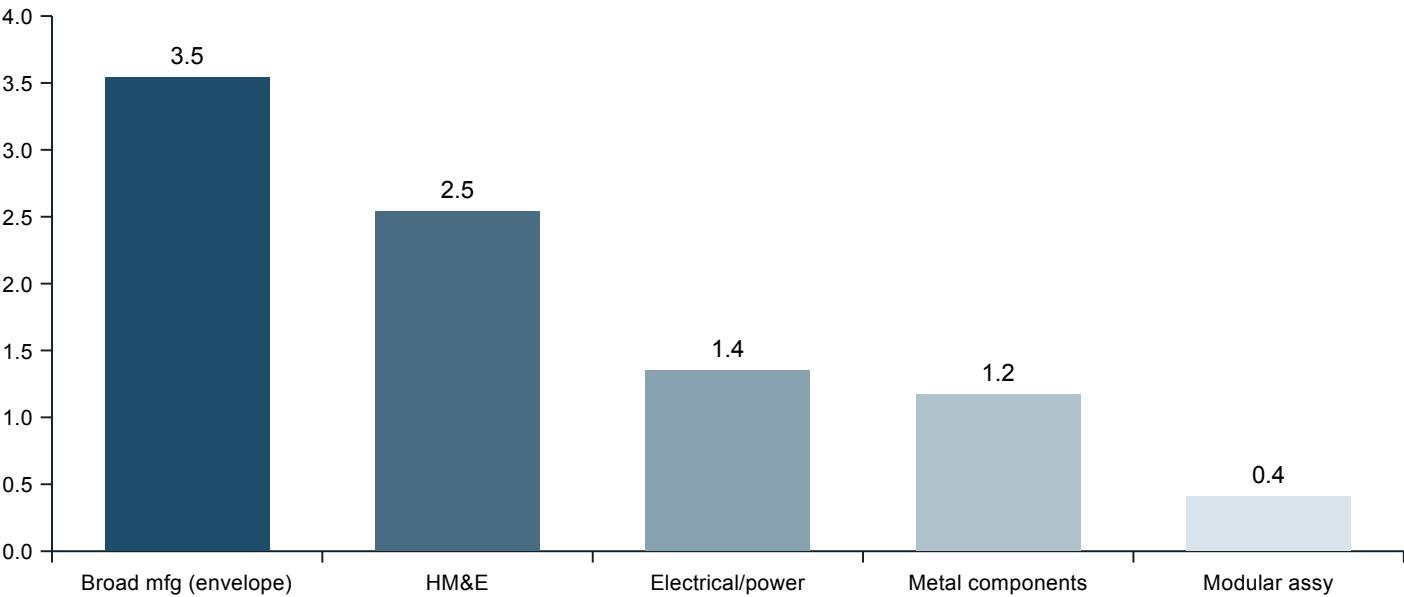


Sources: (1) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026; (2) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (3) parent-prime PIID program-scope attribution



# SAM Scenarios | Broad component manufacturing is the \$3.5B envelope; HM&E is the largest cut at \$2.5B and modular is the smallest at \$408M.

Average annual scenario SAM, \$B, FY2022–FY2027; constant FY2026 dollars. Each scenario is an inclusion cut of one TAM, not additive.



**Scenario composition** **Broad** = all seven buckets; **HM&E** = machining, piping/valves/pumps, electrical/power, and HVAC; **Electrical** = electrical/power only; **Metal** = structural/pre-outfit, machining, and castings/forgings; **Modular** = entity-flagged module-assembly work, not a bucket union

- Broad is the serviceable envelope, not another market.**
- Broad component manufacturing is \$3.5B per year, equal to \$4.2B of TAM less the \$653M residual; every other scenario is a cut of that envelope.
- HM&E is the largest overlapping cut.**
- HM&E reaches \$2.5B because it includes electrical power, machining, piping/valves/pumps, and HVAC; electrical alone is \$1.4B and metal is \$1.2B.
- Modular is smaller but most aligned with distributed build.**
- The \$408M modular cut is entity-flagged module-assembly work, not a structural-and-coatings union.

**Note:** Scenarios are overlapping cuts of one TAM.

Sources: (1) Navy SCN and P-5c Basic Construction and AP/LLTM budget justification, FY2022–FY2027; (2) FFATA/FSRS subaward records and the operating-entity supplier registry; (3) DDG and submarine SAM scenario inclusion-cut definitions





# Evidence and Implications

*Assessing opportunities and the current supplier landscape*



# Supplier Visibility | Visible suppliers prove the layer exists, but first-tier filings capture only a floor.

DDG VISIBLE SUPPLIERS

Leonardo

Arctic Slope

Major Tool

GE

Rolls-Royce

Northrop Grumman

Johnson Controls

SUBMARINE VISIBLE SUPPLIERS

Northrop Grumman

Leonardo

Curtiss-Wright Electro-Mechanical

Scot Forge

DC Fabricators

Rhoads

Graham

Austal

DDG FFATA-VISIBLE YARD-SIDE FLOW

~\$2.7B *vs estimated yard-side outsourcing midpoint ~\$13.6B*

SUBMARINE SUPPLIER-ADDRESSABLE VISIBLE VALUE

~\$5.5B *across ~150 classified recipients; broader visible parent universe ~759*

**Reporting limits:** reporting threshold and lag; yard filing differences; HII Newport News visibility gap

**Supply-chain depth:** direct material booked as direct cost; lower-tier subcontracting; standing agreements

**Visible supplier names prove the layer exists, not the target list.**

- DDG and submarine filings identify real suppliers and work-type cues, but some visible names are services, prime-affiliated, or outside the physical-component boundary.

**First-tier filings set a floor, not the market size.**

- Visibility thins below tier one and varies by yard, so TAM is sized from the budget base rather than by summing subawards.

Sources: (1) FFATA/FSRS subaward records and the operating-entity supplier registry, FY2017–FY2026; (2) DDG MYP-corrected outside-yards place-of-performance evidence; (3) submarine visible-supplier classification and parent-universe analysis

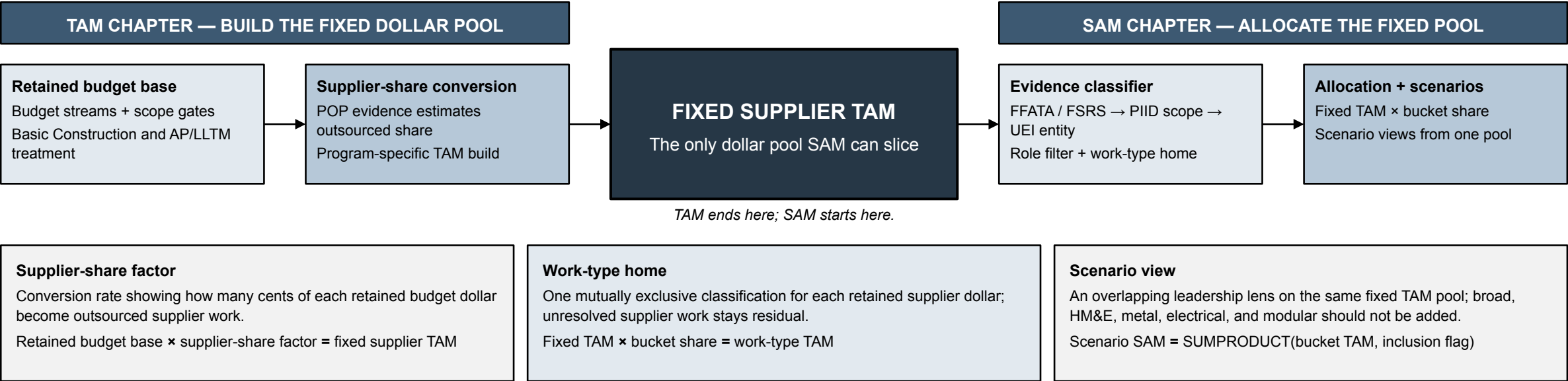


# Appendix



# Methodology (1/5) | TAM creates the fixed supplier pool; SAM slices it without resizing the market.

TAM sizing window FY2022–FY2027; SAM evidence window FY2017–FY2026. Budget and POP evidence size the dollar pool; award evidence allocates the pool.



**TAM and SAM answer different questions.**

- TAM sizes how much outsourced supplier-addressable new-construction work exists; SAM identifies which work-type slices are serviceable inside that fixed pool.

**Award evidence moves allocation, not headline size.**

- PIID, UEI, registry, role filters, residual treatment, and scenario flags can move bucket shares and scenario SAM, but not budget-derived TAM.

**Guardrails:** POP evidence creates supplier-share factors, not a summed award numerator. FFATA / FSRS evidence creates mix percentages, not market size. Scenario views overlap and should not be added.

Sources: (1) Navy SCN budget justification books, FY2022–FY2027; (2) DoD/DoW contract-announcement POP corpus, CY2022–CY2026; (3) SAM.gov FSRS / FFATA subawards and SAM.gov / USAspending UEI / NAICS enrichment, FY2017–FY2026



Methodology (2/5) | Scope gates define the retained budget base before supplier-share conversion.

FY2022–FY2027 constant FY2026 dollars. DDG and submarine are run separately; this page stops at the retained budget stream bases sent to the supplier-share step.

**Retained-base output:** retained stream bases by program: DDG Basic Construction (incl. OBBBA mandatory) + additive AP/LLTM base; submarine Basic Construction + OBBBA mandatory stream + \$0 additive AP/LLTM base. Supplier-share factors are applied next.

Retained-base ledger: what survives into supplier-share conversion

| STEP                         | SOURCE FIELDS USED                                                                                             | RULE APPLIED                                                               | RETAINED OUTPUT                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------|
| 1. Program budget base       | Program; FY; appropriation; SCN / P-5c line; P-10 AP / LLTM / EOQ fields; OBBBA Sec. 20002 mandatory line      | Build the budget base by program and stream                                | Budget is the sizing base; award flow is not           |
| 2. Scope & cost boundary     | Total ship estimate; GFE / GFP; ordnance; plans; nuclear; sustainment / depot; design-only                     | Retain the new-construction component boundary; remove non-component scope | Total ship cost stays context                          |
| 3. Basic Construction stream | P-5c Basic Construction; program flag; FY flag; non-GFE construction base                                      | Treat Basic Construction as the core retained denominator                  | BC retained base by program                            |
| 4. AP/LLTM incrementality    | AP / LLTM / EOQ amount; overlap & incrementality flags; DDG construction share; submarine AP-bridge exclusions | Add only where incremental to Basic Construction                           | DDG additive; submarine nets to \$0                    |
| 5. Handoff to next step      | Retained BC base; retained AP/LLTM base; stream flag; program flag                                             | Send the bases to supplier-share conversion                                | No blended supplier-share factors; no supplier TAM yet |

| PROGRAM TREATMENT: RUN SEPARATELY                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DDG                                                                                                                                                                                                                                        |
| <b>Core base:</b> Basic Construction, LI 2122; OBBBA Sec. 20002(17) mandatory BC folds into the BC stream.<br><b>AP/LLTM:</b> additive, ship-construction-share filtered.<br><b>Output:</b> BC base (incl. OBBBA) + additive AP/LLTM base. |
| SUBMARINE                                                                                                                                                                                                                                  |
| <b>Core base:</b> Virginia + Columbia Basic Construction; OBBBA Sec. 20002(16) runs as its own stream.<br><b>AP/LLTM:</b> reference-only; additive base = \$0.<br><b>Output:</b> BC base + OBBBA stream + \$0 additive AP/LLTM base.       |

- Retained base stays program-specific.**
- DDG and submarine stay separate because AP/LLTM rules and supplier-share evidence differ.
- Conversion happens next.**
- This page sizes eligible retained budget dollars; the next page applies supplier-share factors.

**AP/LLTM incrementality test: add only when not already inside the Basic Construction boundary**

CY AP in window  
**\$1.8B**

×

Ship-construction share  
**80%**

=

Retained AP/LLTM base  
**\$1.5B**

Gross P-10 AP  
**\$44.7B**

–

GFE / design / weapons  
**\$27.3B**

–

Already in BC  
**\$16.8B**

–

Overlap  
**\$0.6B**

=

Additive AP base  
**\$0**

The next step applies the AP/LLTM supplier-share factor; this page stops at the retained base.

AP/LLTM stays useful evidence for supplier-share context, but it is not an incremental base.

Sources: (1) Navy SCN budget justification books, FY2022–FY2027; (2) P-5c Basic Construction and P-10 AP/LLTM budget exhibits; (3) DDG-51, Virginia, and Columbia scope-exclusion reconciliations; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002

# Methodology (3/5) | POP evidence converts retained budget into fixed outsourced supplier TAM.

FY2022–FY2027 retained budget bases from the budget-base step. POP records estimate supplier share; they are not summed as the TAM numerator.

**Supplier-share factor** = conversion rate that turns retained budget dollars into outsourced supplier TAM. It answers: *for every \$1 of retained budget base, how many cents become supplier work?*

How POP evidence becomes the supplier-share factor

Eligible POP dollars  
Confirmed gated in-scope

×

Supplier-location share  
Other US + Foreign POP %

=

Weighted supplier dollars  
Dollar-weighted footprint

+

Eligible POP dollars  
Same gated denominator

=

Supplier-share factor  
Applied to retained base

DDG footprint: *BIW % + Ingalls % excluded; Other US % + Foreign % counted as supplier share.*

| DDG: ADDITIVE AP/LLTM STREAM                            |                      |                                 |                      |                                |
|---------------------------------------------------------|----------------------|---------------------------------|----------------------|--------------------------------|
| BC + OBBBA mandatory base<br>\$21.5B                    | ✖                    | BC supplier-share factor<br>13% | =                    | BC supplier TAM<br>\$2.7B      |
|                                                         |                      |                                 |                      | +                              |
| Retained AP/LLTM base<br>\$1.5B                         | ✖                    | AP/LLTM supplier factor<br>85%  | =                    | AP/LLTM supplier TAM<br>\$1.2B |
| DDG supplier TAM<br>\$4.0B cumulative / \$659M per year |                      |                                 |                      |                                |
| 154 modeled POP actions                                 | \$21.7B gated corpus | \$6.1B excluded / GFE           | \$14.6B MYP restored |                                |

Submarine footprint: *EB % + HII % excluded; Other US % + Foreign % counted as supplier share.*

| SUBMARINE: AP/LLTM REFERENCE-ONLY                                                     |  |                      |                                 |  |                       |                            |  |
|---------------------------------------------------------------------------------------|--|----------------------|---------------------------------|--|-----------------------|----------------------------|--|
| BC + OBBBA mandatory base<br>\$60.6B                                                  |  | ✖                    | BC supplier-share factor<br>35% |  | =                     | BC supplier TAM<br>\$21.2B |  |
|                                                                                       |  |                      |                                 |  |                       |                            |  |
| AP/LLTM additive base<br>\$0                                                          |  | ✖                    | Reference factor<br>48%         |  | =                     | Additive TAM<br>\$0        |  |
|                                                                                       |  |                      |                                 |  |                       |                            |  |
| AP/LLTM is supplier evidence, not incremental TAM: already inside BC or out of scope. |  |                      |                                 |  |                       |                            |  |
| Submarine supplier TAM<br>\$21.2B cumulative / \$3.5B per year                        |  |                      |                                 |  |                       |                            |  |
| 658 POP rows                                                                          |  | \$25.4B gated corpus |                                 |  | \$6.1B excluded / GFE |                            |  |

FIXED OUTSOURCED SUPPLIER TAM

\$25.2B cumulative / \$4.2B per year

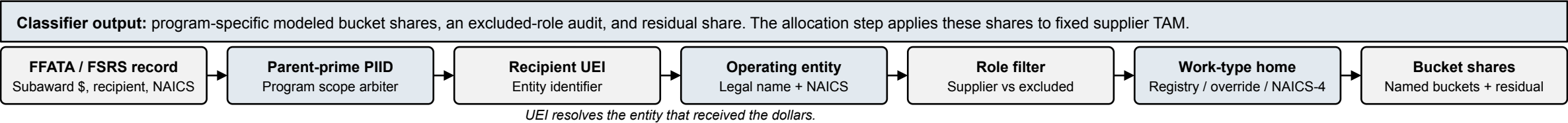
This is the fixed dollar pool the allocation step will slice; SAM does not resize it.

- The supplier-share factor is a conversion rate, not an award total.
- POP records estimate the supplier share of retained budget dollars; the Navy budget base still supplies the TAM dollars.
- DDG and submarine stay separate until outputs are summed.
- The two programs have different yard footprints and AP/LLTM treatment, so the model sums TAM dollars rather than blending supplier-share factors.

Sources: (1) DoD/DoW contract-announcement POP corpus, CY2022–CY2026; (2) FPDS Atom feeds and DDG MYP master-total reconstruction; (3) Navy SCN P-5c / P-10 budget exhibits, FY2022–FY2027; (4) FY 2026 Mandatory Funding Allocation Plan, PL 119-21 Sec. 20002

# Methodology (4/5) | PIID scopes the award, UEI resolves the operating entity, and the registry assigns one work-type home.

FY2017–FY2026 FFATA / FSRS evidence window. Award records set supplier mix percentages; the fixed TAM dollar pool comes from the supplier-share step.



Classification ledger: how a record becomes one work-type home

| STAGE                  | SOURCE FIELDS USED                       | RULE APPLIED                      | OUTPUT / GUARDRAIL        |
|------------------------|------------------------------------------|-----------------------------------|---------------------------|
| 1. Award pull          | Subaward ID, parent PIID, UEI, NAICS, \$ | Collect visible supplier evidence | Mix, not TAM              |
| 2. PIID scope          | Parent-prime + discovered PIID flags     | Scope to DDG or submarine         | Not hull-level proof      |
| 3. Entity resolution   | sub_entity_uei, legal name, NAICS        | Resolve the operating entity paid | Not parent brand          |
| 4. Role filter         | Role, prime / co-prime, GFE, FMS flags   | Drop non-supplier / non-component | Excluded roles leave base |
| 5. Work-type assign    | Registry bucket, override, NAICS-4       | First clean match wins one home   | QA, not classifier        |
| 6. Residual discipline | Addressable, bucketed, unbucketed \$     | Keep unresolved \$ in denominator | Dilutes named shares      |
| 7. Share output        | Bucket \$ over addressable base          | Modeled work-type and residual %  | Feeds allocation step     |

CLASSIFICATION PRECEDENCE: FIRST CLEAN MATCH WINS

1. Registry / operating-entity UEI assigns role and bucket.

2. Prime / co-prime name excludes final-assembly yards.

3. GFE / SIB / Navy-directed name routes out of supplier SAM.

4. Vendor-name override handles known exceptions.

5. NAICS-4 fallback assigns a bucket where registry is missing.

6. Service / non-component NAICS routes to excluded service.

7. Holding-company NAICS 5511 routes to excluded holding.

8. Foreign / FMS routes out of U.S. supplier SAM.

9. Residual stays supplier / unbucketed when unresolved.

Work-type homes: one per retained supplier dollar

Structural / modules

Machining / propulsion

Castings / forgings

Piping / fluid handling

Electrical power

HVAC / chilled water

Coatings / insulation

Residual / unbucketed

The classifier is entity-first and role-first.

- PIID scopes; UEI resolves the entity; roles filtered first.

Residual is unclassified award spend.

- Unresolved dollars stay in the denominator, outside SAM.

| DDG EVIDENCE BASE           | SUBMARINE EVIDENCE BASE     |
|-----------------------------|-----------------------------|
| 22,867 FFATA / FSRS records | 929 classified entity rows  |
| 118 discovered PIIDs        | \$6.1B all-recipient base   |
| 1,555 cleaned vendors       | \$5.5B supplier-addressable |
| 151 NAICS lookups           | \$664.8M unbucketed value   |
| \$6.0B supplier-addressable | 12% residual / unbucketed   |

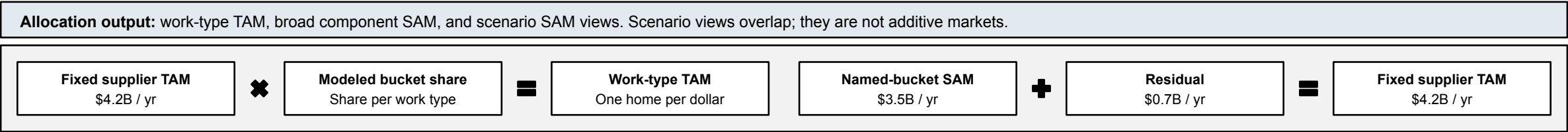
Examples: why operating entity beats parent brand

|                                  |                               |                         |
|----------------------------------|-------------------------------|-------------------------|
| Northrop Sunnyvale generators    | Ship-power operating entity   | Electrical power        |
| Curtiss-Wright / Circor valves   | Industrial fluid-handling     | Piping / fluid handling |
| Lockheed / Raytheon Aegis, SPY-6 | Navy-directed mission systems | Excluded role           |
| Blank NAICS, no registry match   | Supplier evidence, unresolved | Residual / unbucketed   |

Sources: (1) SAM.gov FSRS / FFATA first-tier subaward records, FY2017–FY2026; (2) SAM.gov Entity API and USAspending UEI / NAICS enrichment; (3) FPDS / DoD PIID records and program scope-exclusion registers

# Methodology (5/5) | Fixed TAM is sliced into work-type homes and overlapping scenario views.

FY2022–FY2027 average annual TAM and SAM; constant FY2026 dollars. The supplier-share step supplies the fixed TAM dollar pool; the evidence classifier supplies modeled work-type shares and residual.

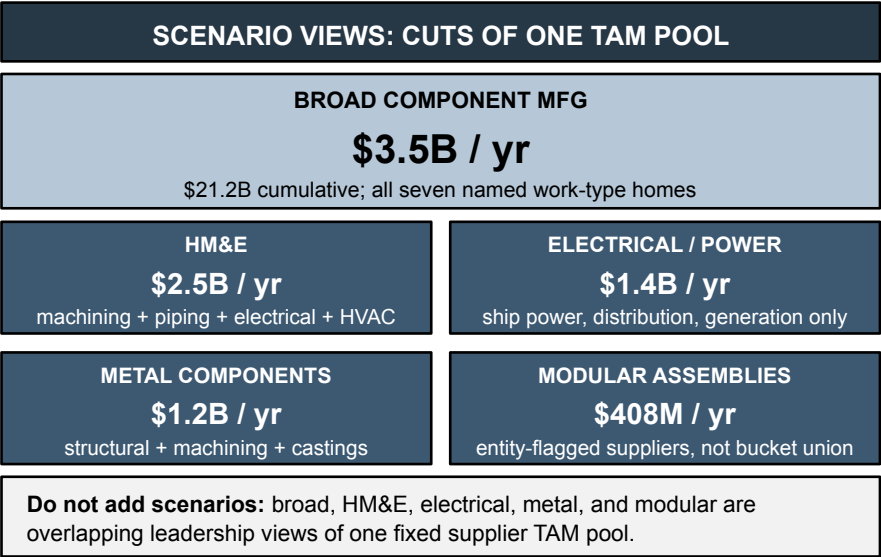


Work-type allocation: mutually exclusive homes mapped into overlapping scenario views

| WORK-TYPE HOME                               | AVG TAM (\$M) | BROAD | HM&E | METAL | ELECTRICAL |
|----------------------------------------------|---------------|-------|------|-------|------------|
| Electrical power / distribution / generation | \$1,353M      | Yes   | Yes  | No    | Yes        |
| Piping / valves / pumps                      | \$710M        | Yes   | Yes  | No    | No         |
| Structural fabrication / modules             | \$633M        | Yes   | No   | Yes   | No         |
| Machining / mechanical / propulsion          | \$375M        | Yes   | Yes  | Yes   | No         |
| Coatings / insulation / decking              | \$206M        | Yes   | No   | No    | No         |
| Castings / forgings                          | \$162M        | Yes   | No   | Yes   | No         |
| HVAC / ventilation / chilled water           | \$102M        | Yes   | Yes  | No    | No         |
| Residual / unbucketed                        | \$653M        | No    | No   | No    | No         |

Residual treatment: residual stays inside fixed TAM and dilutes named-bucket shares, but stays outside broad and targeted scenario SAM.

Modular treatment: modular assemblies come from registry entity flags, not from summing one or more work-type rows.



Work types classify dollars once; scenarios re-cut them multiple ways.

- Each supplier dollar is assigned one work-type home or residual, but Broad, HM&E, Metal, Electrical, and Modular are overlapping views built from those homes and flags.

Residual keeps the allocation conservative.

- Unresolved supplier dollars remain in the fixed TAM reconciliation, but they stay outside named scenario SAM until registry, NAICS, or override evidence supports assignment.

Sources: (1) SAM.gov FSRS / FFATA subaward records, FY2017–FY2026; (2) SAM.gov Entity API / USAspending NAICS enrichment and work-type evidence; (3) DoD/DoW POP corpus and Navy SCN budget books supporting fixed supplier TAM

